

Summary

Software Engineer with 1 year of industry experience building production-grade Python applications, backend services, machine learning pipelines, and LLM-powered systems. Experienced in designing end-to-end AI workflows, integrating external APIs, and deploying scalable applications with measurable business impact across forecasting, decision intelligence, and real-time analytics.

Professional Experience

Data Scientist

Arabania Indo-Arabic Restaurant

June 2024 - May 2025, Lucknow

- Developed and deployed time-series forecasting models using Python (statsmodels, scikit-learn), reducing food waste by 20% and procurement costs by 12% across 100K+ inventory records.
- Designed and executed end-to-end A/B pricing experiments using statistical hypothesis testing, identifying pricing strategies that improved gross margins by 15% within two months.
- Engineered supervised machine learning models using feature engineering on transactional and behavioral data to predict SKU performance, improving operational efficiency by 12%.
- Developed automated Python and SQL data pipelines leveraging CTEs and window functions for demand analytics and reporting, reducing manual reporting effort by 40%.

Projects

European Power Market Forecasting System

GitHub

Python, FastAPI, XGBoost, ENTSO-E API, LLM Integration, Prompt Engineering

- Architected modular backend services for electricity price forecasting, covering data ingestion, validation, feature engineering, model inference, trading signal generation, and PnL simulation.
- Engineered an automated data quality framework performing timestamp validation, anomaly detection, statistical consistency checks, and economic sanity validation before model execution.
- Integrated ENTSO-E and LLM APIs using structured prompt engineering to automatically generate trader summaries and actionable market insights, reducing manual reporting by 3+ hours per week.

AI Customer Churn Decision Intelligence Platform

Live App

Python, FastAPI, REST APIs, Scikit-learn, LLM APIs, Streamlit

- Designed and deployed a production-ready FastAPI backend exposing REST APIs for customer analysis, scenario simulation, and real-time prediction through an interactive Streamlit application.
- Designed a modular AI workflow combining machine learning inference, workflow orchestration, LLM provider abstraction, and structured prompt engineering to deliver personalized retention recommendations.
- Implemented batch inference, input validation, environment-based configuration, and cloud deployment on Render and Streamlit Cloud to support scalable real-time customer analysis.

Real-Time Urban Analytics Dashboard

Live App

Python, REST APIs, Streamlit, PyDeck, Forecasting

- Developed a modular Python application integrating live traffic, weather, and air quality data from multiple REST APIs with centralized configuration management and reusable service modules.
- Designed automated data ingestion, preprocessing, caching, and alert pipelines with API response validation to support reliable real-time analytics.
- Built interactive monitoring dashboards with forecasting, anomaly detection, and geospatial visualization for simultaneous multi-city operational insights.

Technical Skills

- **Backend Engineering:** Python, FastAPI, REST APIs, SQL, Database Design, Service Integration
- **Cloud & DevOps:** Docker, AWS (EC2, Lambda), Kubernetes, CI/CD Fundamentals, Render, Streamlit Cloud
- **LLMs & AI Systems:** OpenAI API, Anthropic API, Groq API, Prompt Engineering, Custom AI Agents, Workflow Orchestration, Structured Output Validation
- **Machine Learning & Data:** Pandas, NumPy, scikit-learn, XGBoost, Time-Series Forecasting, Feature Engineering, Automated Data Pipelines, Git, Vue.js

Certifications

- Google Data Analytics Professional Certificate

Education

Indian Institute of Technology, Madras

2022-2025

B.Sc in Programming and Data Science